Amrita Kaur

EV Systems Engineer

RESUME_1776

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Total Experience: 12 years

PROFESSIONAL SUMMARY

EV Systems Engineer with 12 years of experience in ev systems, using Battery Management Systems, CAN Bus, MATLAB, Simulink. Experienced in model-based design, automotive safety and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: MATLAB, Motor Control, HIL, SOC Estimation, Python, Battery Management Systems, Simulink, EV Powertrain, CAN Bus, Diagnostics, AUTOSAR, ISO 26262

Methods: Model-based design, Automotive safety, Vehicle networks

PROFESSIONAL EXPERIENCE

Lead EV Systems Engineer | VegaBridge Systems | Jan 2022 - Present

- Validated 83 HIL fault-injection scenarios for protective responses
- Lowered peak powertrain losses by 27% in control-model experiments
- Reduced simulated fault-detection delay by 33% using diagnostic timing refinements
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

EV Systems Engineer | ZenithTrail Technologies | Jan 2018 - Dec 2021

- Reduced simulated fault-detection delay by 25% using diagnostic timing refinements
- Improved SOC estimation error by 20% after parameter calibration

Associate EV Systems Engineer | SparrowCloud Labs | Jul 2014 - Dec 2017

- Validated 769 HIL fault-injection scenarios for protective responses
- Lowered peak powertrain losses by 40% in control-model experiments

PROJECTS

Charging Control Study | HIL, EV Powertrain, Simulink, ISO 26262

- Simulated voltage and current control under load variations.
- Documented acceptance criteria and evaluated the design using automotive safety and vehicle networks.

Battery Pack Health Monitor | HIL, Diagnostics, Battery Management Systems, AUTOSAR

- Implemented SOC estimation checks and temperature fault diagnostics.
- Documented acceptance criteria and evaluated the design using automotive safety and vehicle networks.

EDUCATION

B.Tech in Electrical Engineering, Narmada School of Technology, 2014

Relevant Methods: Model-based design, Automotive safety, Vehicle networks

CERTIFICATIONS

Battery Management Workshop (2025)